

eclipseBord - FastlyDep

Purpose of this lab is to use your knowledge in Python, FastAPI, Streamlit as well as Docker and Azure to build fullstack application that you will deploy. You will start with building the solution locally, then dockerize it and then deploy it to Azure.



Figure 1: image generated from nano banana 2

Background

You work at a company called **FastlyDep**, and if you haven't noticed yet the name has the componenets FastAPI + Streamlit + Deploy in it. They know that you also have worked with this stack and that you also love data.

As you might have noticed, there was a very cool event happening 12th august 2026 - no I am not talking about way out west. I am talking about the solar eclipse, where the moon covers the sun. Of course it would be fun to create a dashboard about the solar eclipse data. FastlyDep wants you to create eclipseBord and deploy it to Azure.

Tasks

FastlyDep has divided it into several subtasks for you to complete

Task 0 - Warmup and prerequisites

Go into this [github repository](#), watch the videos part 1 to 3 in *15_dockerize_deploy_fastapi_streamlit*, to learn about uv workspaces, fastapi, streamlit and deploying to Azure.

Create a new public github repository for your eClipseBord project.

Download the [dataset from here](#).

Task 1 - EDA

Create a short EDA of the dataset to understand it briefly. Don't overmake the EDA here, the focus is on cloud infrastructure and not on data itself in this project.

Task 2 - Build the application and deploy

Build the application and deploy it

- setup uv workspaces
- build backend
- build frontend
- dockerize
- deploy to Azure

Make the deployment on the portal first and if you have time over, you can do it with Terraform.

Task 3 - Video

Create a 5-10 min long video where you showcase your **deployed** dashboard, and explain the architecture - both locally, docker and azure infrastructure.

Simplest would be to record your video with Teams and then create an unlisted youtube video and then send the link in submission.

Submission

Hand in the following to your learning platform:

- a link to your public github repository for this lab
- link to your video

LLM usage

LLMs for coding are allowed, for smaller parts, for generating ideas, but not for solving entire parts. Very important is that you understand the code, the hardware and your product. If there are parts where you use LLMs for coding, it's important that you make a comment that it is LLM generated.

LLM should not be used in the individual report

Grading

There is no VG on this lab

Criteria for G

- fulfilled all tasks
- can explain what you have done in the video